

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Regular and Supplementary Winter Examination – 2024

Course: B. Tech.

Branch : Electronics Engineering/ Electronics & Telecommunication Engineering / Electronics and Communication Engineering

Semester : VII

Subject Code & Name: BTEXOE704C/ BTETOE704C & Data Structure & Algorithms Using Java Programming

Max Marks: 60

Date:12/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 What is a data structure?	L1	1
a. A programming language		
b. A collection of algorithms		
c. A way to store and organize data		
d. A type of computer hardware		
2 Which of the following is an example of an Elementary Item?	L1	1
a. Student Name		
b. Student Roll Number		
c. Student Age		
d. Student Address		
3 The time complexity of Linear Search in the worst case is:	L1	1
a. $O(n)$		
b. $O(\log n)$		
c. $O(n^2)$		
d. $O(1)$		
4 In Binary Search, how do you determine the next search interval?	L1	1
a. Compare the middle element with the target and choose either the left or right half		
b. Compare all elements sequentially		
c. Divide the array into three equal parts		
d. Perform a linear scan to find the next interval		
5 What is the primary principle followed by a Stack?	L2	1
a. FIFO (First In, First Out)		
b. LIFO (Last In, First Out)		
c. Random Order		
d. FILO (First In, Last Out)		
6 Which operation checks the topmost element of a stack without removing it?	L3	1
a. Pop		
b. Push		
c. Peek		
d. Enqueue		
7 Which of the following is NOT a type of linked list?	L3	1
a. Singly Linked List		
b. Doubly Linked List		
c. Circular Linked List		
d. Static Linked List		
8 In a Doubly Linked List, how many pointers does each node have?	L2	1
a. One pointer to the next node		
b. Two pointers: one to the next node and one to the previous node		
c. Three pointers: next, previous, and parent		
d. No pointers		
9 What is a leaf node in a tree?	L1	1
a. A node with no parent		
b. A node with exactly one child		
c. A node with no children		
d. A node with multiple parents		

10	In a binary tree, what is the maximum number of children a node can have?	L1	1
	a. 1 b. 2 c. 3 d. Unlimited		
11	What is the time complexity to access an element in an array using an index?	L3	1
	a. $O(n)$ b. $O(\log n)$ c. $O(1)$ d. $O(n^2)$		
12	If an array is declared as int arr[10] ; how many elements can it store?	L2	1
	a. 10 b. 11 c. 9 d. 8		
Q. 2	Solve the following.		12
A)	Define Searching, Explain Linear search method with the help of example; also give best, worst, average case of the same.	L1	6
B)	List different operations performed on Data structure. Explain any two with example.	L2	6
Q.3	Solve the following.		12
A)	Convert the infix expression $(A+B)*C/D$ to	L3	6
	1) Prefix expression		
	2) Postfix expression		
B)	Explain Stack data structure. Write an algorithm of PUSH and POP operations of stack.	L2	6
Q. 4	Solve Any Two of the following.		12
A)	Write an algorithm to insert node at the beginning in Single Linked list.	L3	6
B)	Write differences between array and linked list.	L1	6
C)	Explain representation of linked lists in the memory.	L2	6
Q.5	Solve Any Two of the following.		12
A)	What is a binary tree? Explain any two types of binary tree with an example.	L1	6
B)	Define the following terms with neat diagrams:	L2	6
	1) Siblings		
	2) Leaf Node		
	3) Ancestor of Node		
C)	Write Binary Tree Properties.	L2	6
Q. 6	Solve Any Two of the following.		12
A)	How does the Bubble sort works? Explain with the help of example.	L1	6
B)	Write an algorithm for Insertion sort.	L3	6
C)	What is a Graph? Explain the following terms	L2	6
	1) Undirected graph		
	2) Directed graph		
	3) Weighted and Unweighted graph		
	*** End ***		